

Algebra/Topology Seminar

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NIELSEN REALIZATION FOR OUTER AUTOMORPHISMS OF 3-MANIFOLD GROUPS

Thursday, October 15, 2026

3:00 p.m. in Massry B008

ABSTRACT. For any smooth, closed, orientable manifold M , the mapping class group $\text{Mod}(M)$ is the group of orientation-preserving diffeomorphisms of M , up to isotopy. The Nielsen realization problem asks which finite subgroups of $\text{Mod}(M)$ can be lifted to finite groups of diffeomorphisms. In this talk we will study a variant of this problem, in which we investigate lifting finite subgroups of the outer automorphism group of the fundamental group of M to $\text{Mod}(M)$. In most cases, we find a complete set of cohomological obstructions that determine whether such a lift exists. This is joint work with B. Tshishiku.